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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	TimeOut
Selected Domain	Sustainable food manufacturing
Selected SDG Track (1–17)	SDG 8 Decent Work and Economic Growth
Project / Solution Name	Timed Production Planning
Name (Leader)	Muhammad Husein Ilhamsyah
Email & Phone (Leader)	Husein.Ilhamsyah@gmail.com
University Affiliation(s)	Sunway University
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small food manufacturers must coordinate customer deadlines, ingredient stocks and limited production hours. Seasonal demand can change quickly, while a first-come, first-served schedule can consume scarce ingredients before urgent orders are reached. Owners need an affordable way to see what can be produced, which materials are short and when to reorder. Timed demonstrates this challenge using Kuih Mesra Sdn Bhd, a fictional Malaysian food business. In the synthetic six-order example, butter stock is 15 kg against 19.2 kg needed for cookie orders, and chicken stock is 25 kg against 35 kg required. First-come, first-served scheduling completes only three orders on time. Poor planning can contribute to missed sales, rushed purchasing and food loss through mismatched production. The prototype addresses this planning problem; it does not yet measure actual spoilage or environmental savings. Its intended users are owners and production staff in small food businesses.	

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Timed combines weekly demand forecasting, ingredient checks and deadline-based scheduling. A custom regression learns patterns from sales history, prices, promotions, school holidays, Ramadan and festivals, with outlier filtering and estimated 80% forecast ranges. The planner checks stock, recipes and working hours, then prioritises earlier due dates. Its synthetic demonstration improves on-time completion from three to four of six orders. This heuristic does not guarantee a globally optimal schedule. Stockout alerts use supplier lead times, and purchase requests require owner approval. Optional language-model APIs explain calculated facts in English or Bahasa Melayu; built-in English briefings work offline. API integration, SQLite storage and a web interface provide a basis for applying GenAI, database and cloud workshop topics. Workshop attendance requires confirmation. Phase 2 will deploy the prototype and test usability, planning performance and food-waste outcomes.	

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16.4.2, React 19.3.0, TypeScript and Tailwind CSS. Backend: Python, FastAPI, Uvicorn and Pydantic. Forecasting: NumPy and pandas; CSV/Excel exchange: pandas and openpyxl. Storage: SQLite for the current dataset, purchase orders and activity log. Flow: user -> Next.js interface -> FastAPI -> validated production data -> forecast and stockout analysis plus scheduling -> metrics and purchase requests -> SQLite and CSV/Excel exports. Optional calculated facts -> configured LLM API -> checked explanation -> user. Groq, Gemini and OpenRouter adapters are implemented as optional alternatives. Their actual use and account tiers require team confirmation. No sponsor API or cloud deployment is established by the project files. Frontend data connections currently use localhost; Phase 2 requires deployment configuration and secure access controls.	

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The prototype reflects an owner-and-staff workflow: inspect orders and inventory, compare plans, understand shortages, and retain owner approval for purchases. Its synthetic scenario makes the planning trade-off visible before collecting business data. Stakeholder interviews and usability testing are planned; completed user research is not claimed. Phase 2 will validate the workflow with a small food business, test CSV/Excel onboarding and compare schedules against the same baseline inputs. The team will deploy the web frontend and Python API, configure a persistent database volume, replace localhost URLs and add authenticated roles, HTTPS and a retention policy. Deployment services remain to be selected. Acceptance checks will cover reproducible forecasts, inventory feasibility, owner-only approval, export consistency, and useful explanations when an AI provider is unavailable. A measured pilot will record on-time fulfilment and discarded food before making environmental impact claims.	

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 12.3: reduce food losses along production and supply chains. Supporting target 12.2: use natural resources efficiently. Timed aims to improve planning and ingredient use. Source: UN DESA, Goal 12, https://sdgs.un.org/goals/goal12 (accessed 25 October 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Synthetic baseline: 3/6 orders on time (50%); 3/6 infeasible. Butter shortage: 4.2 kg; chicken shortage: 10 kg. Sources: data/README.md, materials.csv, orders.csv and recipes.csv; reproduced with core.metrics.compare_plans. Actual food-waste baseline will be measured during the pilot.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary environmental KPI: discarded edible food kg per 100 kg of ingredients processed. Operational KPI: on-time orders / total orders, compared on identical inputs. Supporting KPI: forecast MAPE on a chronological holdout. Log disposal, inputs and fulfilment during the pilot.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed targets: 10% lower food-waste intensity against a measured pilot baseline; 20% relative improvement in on-time fulfilment against first-come, first-served. Synthetic demonstration: 50% to 66.7% on time, a 33.3% relative gain. Actual waste reduction is not yet measured.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, Tailwind; FastAPI, Pydantic, HTTPX, openpyxl	Third-party libraries	MIT except HTTPX: BSD-3-Clause
NumPy, pandas, Uvicorn; TypeScript; pytest; python-multipart	Third-party libraries	BSD-3-Clause; Apache-2.0; MIT; Apache-2.0 respectively
Demo data; custom model; optional LLMs	Dataset and models	Repo Apache-2.0; external provider terms; team to confirm use

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Muhammad Husein Ilhamsyah — Whoosayinn

25 October 2026